

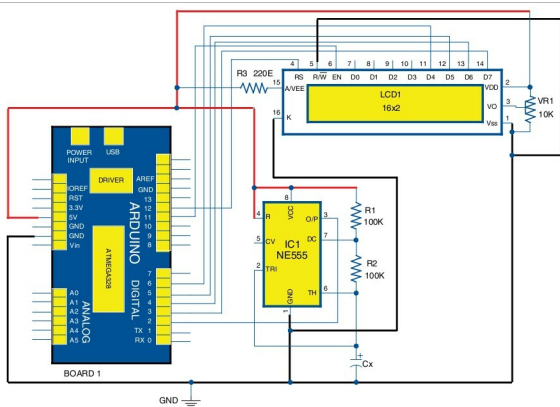


Fig. 3: Circuit diagram of Arduino based digital capacitance meter with NE555 timer in astable mode

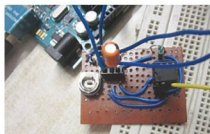


Fig. 4: NE555 timer connection in author's breakout board

to GND. A 10-kilo-ohm potentiometer (VR1) is provided for adjusting the contrast of the display.

Software

The source code (capacitance1.ino) is written in Arduino programming language. Atmega328P/Atmega328P is programmed using Arduino IDE software. Select the correct board from Tools→Board menu in Arduino IDE and burn the program (sketch) through the standard USB port in your computer.

Here, code written in Arduino

uses LiquidCrystal.h header file provided by Arduino library for working with the LCD.

lcd.begin(16, 2) function helps configure the 16 × 2 character LCD.

Serial.begin(9600) function initialises the serial port with a baud rate of 9600.

attachInterrupt(0, cap, RISING) function calls interrupt handler 'cap' whenever a signal connected to interrupt 0 pin (pin 2) of Arduino makes a low-to-high, that is, rising-edge transition.

Serial.print(capacitance, 3) function prints the measured value of capacitance up to three decimal places using the serial port on the PC monitor.

lcd.setCursor(0, 1) function sets the LCD cursor position to print from first column of second row.

lcd.print(capacitance, 3) function prints the measured value of capacitance up to three decimal places on the LCD screen.

NE555 timer in monostable mode

In the second method (Fig. 5), NE555 timer (IC1) is operated in monostable multivibrator mode with external resistor R1 and an unknown capacitor (Cx), whose value is to be measured. IC1 is powered with +5V from Arduino board (Board 1); thus, eliminating the need for an external DC power supply.

In this mode of operation, when trigger pin of the timer is made low (0V) by sending an active low pulse from pin 9, output of the timer (from pin 3) goes high (+5V) for a certain period of time, which is determined by the values of R1 and Cx. Time period (T) for which the timer's output remains high is given as:

$$T = 1.1 \times R1 \times Cx$$

Output of the timer is connected to pin 2 of Arduino, which is a hardware interrupt pin (called Interrupt 0). The source code (capacitance2.